

StudyStack

Product Requirements Document

Client-Ready Version

A subject-agnostic learning platform for syllabus tracking, AI-powered doubt resolution, and previous-year paper analysis.

1. Executive Summary

StudyStack is a subject-agnostic web platform designed for college students across academic streams. It combines syllabus tracking, AI-powered doubt resolution, and previous-year question paper analysis in one application. The platform helps students understand what to study, monitor their progress, resolve conceptual doubts, and identify high-priority exam topics.

The initial release will focus on a deployable minimum viable product (MVP) with an estimated AI operating cost of \$5-\$10 per month.

2. Problem Statement

- Students do not have a unified tool for tracking syllabus progress across subjects.
- Students outside coaching institutes often lack real-time, contextual support for resolving doubts.
- Analyzing previous-year question papers manually to identify recurring topics is time-consuming and inconsistent.
- Existing solutions such as chat applications, videos, and static PDFs are fragmented and are not personalized to each student.

3. Objectives

- Enable topic-level syllabus tracking with clear progress visibility.
- Provide AI-based doubt resolution through text and image input.
- Automate previous-year paper analysis to identify frequently asked topics.
- Provide a unified dashboard showing progress, doubts, and high-frequency topics.
- Deliver a subject-agnostic MVP within an estimated AI operating cost of \$5-\$10 per month.

4. Target Users

User Persona	Primary Need
University student (any stream)	Track syllabus progress and resolve doubts quickly
Self-studying student	Understand examination patterns and prioritize study topics
Last-minute exam preparation student	Receive a list of high-probability topics for revision

5. Scope

5.1 In Scope for Version 1

- User authentication.
- Subject and topic creation with progress tracking.
- AI doubt solver with text and image input.
- Previous-year paper upload and topic-frequency analysis.
- Unified dashboard for progress, doubts, and high-frequency topics.

5.2 Out of Scope for Version 1

- Multi-user classrooms and teacher dashboards.

- Payment or subscription functionality.
- Native mobile applications.
- Real-time collaborative study rooms.
- Automatically generated mock tests.

6. Functional Requirements

6.1 Syllabus Tracker

ID	Requirement	Priority
FR-1.1	The user can create subjects and optionally add an examination date.	Must
FR-1.2	The user can add topics using free-form text input.	Must
FR-1.3	Each topic can have one of three statuses: Not Started, In Progress, or Done.	Must
FR-1.4	The system calculates a progress percentage for each subject.	Must
FR-1.5	The user can manually mark a topic as weak.	Should
FR-1.6	The system can display a days-to-examination countdown.	Could

6.2 AI Doubt Solver

ID	Requirement	Priority
FR-2.1	The user can submit a text-based doubt within a selected subject.	Must
FR-2.2	The user can upload handwritten or printed questions as images.	Must
FR-2.3	The system extracts text from uploaded images using OCR.	Must
FR-2.4	The system generates a step-by-step explanation using an LLM.	Must
FR-2.5	The system stores chat history for each subject.	Must
FR-2.6	The system applies a daily AI-call limit per user.	Must

6.3 Previous-Year Paper Analyzer

ID	Requirement	Priority
FR-3.1	The user can upload previous-year papers in PDF format.	Must
FR-3.2	The system extracts text and separates the paper into individual questions.	Must
FR-3.3	The system classifies each question by topic using an LLM.	Must
FR-3.4	The system matches questions to syllabus topics using fuzzy matching and LLM support.	Must
FR-3.5	The system generates a frequency table showing topics and the years in which they appeared.	Must
FR-3.6	The system ranks topics according to their estimated examination probability.	Must

7. Non-Functional Requirements

Category	Requirement
Performance	Text-based doubt responses should be returned within five seconds.
Processing	Paper analysis should run asynchronously and display a processing-status indicator.
Security	User data must be isolated by account, and uploaded files must be accessible only to their owners.

Category	Requirement
Scalability	The backend should be stateless and capable of horizontal scaling.
Cost	AI usage should be controlled through rate limits and batched LLM calls, targeting \$5-\$10 per month.
Availability	A publicly accessible deployed instance should be available for demonstrations.
Usability	A user should be able to create a subject and add the first topic in under two minutes.

8. System Architecture

Layer	Technology
Frontend	Next.js, TypeScript, and Tailwind CSS
Backend - authentication and CRUD	Node.js / Next.js API routes
Backend - AI and PDF processing	Python and FastAPI
Database and authentication	Supabase with PostgreSQL
OCR	Tesseract.js
PDF parsing	pdfplumber / PyPDF2
LLM	Gemini 1.5 Flash / GPT-4o-mini
Deployment	Vercel for the frontend and API; Render or Railway for the FastAPI service

9. Data Model

Entity	Fields
Users	id, email, created_at
Subjects	id, user_id, name, exam_date
Topics	id, subject_id, title, status, is_weak
Doubts	id, subject_id, user_id, question_text, image_url, answer_text, created_at
Papers	id, subject_id, year, file_url, uploaded_at
Paper Questions	id, paper_id, question_text, matched_topic_id, raw_topic_label

10. API Endpoints

Method	Endpoint	Purpose
POST	/auth/signup	Register a new user
POST	/auth/login	Authenticate a user
POST	/subjects	Create a subject
GET	/subjects/:id	Retrieve a subject and its topics
PATCH	/topics/:id	Update a topic status
POST	/doubt-solve	Submit a text or image doubt and receive an AI explanation
GET	/doubts/:subject_id	Retrieve doubt history for a subject
POST	/papers/upload	Upload a previous-year paper
GET	/papers/:subject_id/analysis	Retrieve topic-frequency analysis

11. Assumptions and Constraints

- Users will manually enter syllabus topics; the system will not automatically retrieve syllabi from university sources in Version 1.
- Paper analysis is limited to text-based PDFs. Scanned or image-only PDFs are excluded from Version 1.
- OCR accuracy for handwritten input is not expected to exceed 80%.
- The AI budget is fixed at \$5-\$10 per month and will be controlled through usage limits.
- No fixed delivery timeline has been defined; development will be managed through phase gates.

12. Risks and Mitigations

Risk	Mitigation
Low OCR accuracy for handwritten questions	Prioritize printed or typed input in Version 1 and clearly communicate the limitation.
Incorrect topic classification	Use fuzzy matching as the first pass and LLM classification as a fallback.
Unsupported scanned or image-only PDFs	Document the limitation and restrict Version 1 to text-based PDFs.
AI cost overruns under public usage	Apply per-user daily limits, billing caps, and batched LLM calls.

13. Success Metrics

- Validate the end-to-end workflow across two to three different subjects or academic streams.
- Achieve at least 90% doubt-solver response accuracy during quality-assurance testing.
- Correctly parse at least three distinct previous-year paper formats.
- Provide a deployed, publicly accessible instance with a working demonstration.

14. MVP Delivery Summary

The MVP is considered ready for demonstration when authentication, syllabus tracking, AI doubt resolution, previous-year paper analysis, and the unified dashboard are implemented, tested against the success metrics, and deployed as a publicly accessible application.